A Multiscale Self-Attention Deep Clustering for Change Detection in SAR Images

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***Abstract*— Synthetic aperture radar (SAR) image change detection (CD) is an important application in the field of remote sensing. Due to the lack of labeled data especially in the pixelwise task, it is urgent to develop unsupervised techniques to effectively detect changes. In this article, we propose a novel unsupervised representation learning framework for CD in SAR images, called multiscale self-attention (SA) deep clustering based on octave convolution. The main motivation is that a convolutional neural network (CNN) has the ability to extract significant feature hidden in input images, but it relies heavily on annotated data. Clustering is typically free from supervision; however, SAR images always suffer from speckle noise, which is unfriendly for clustering. Thus, we integrate unsupervised clustering with CNN to learn clustering-friendly feature representations. In the unified framework, CNN feature learning and clustering can be optimized end-to-end without supervision. To better suppress speckle noise and boost the joint optimization for distinguishing changes and unchanges, we use the K-means++ algorithm that is robust to noise as the clustering algorithm. In the meanwhile, we introduce the octave convolution and SA mechanism into the network to fully mine important spatial structure information for enhancing noise resistance of the network. Moreover, multiscale fusion modules are proposed to fuse multiscale input into a complementary feature representation that contains more context and semantic information around each pixel so that it refines the difference feature extraction while reducing speckle noise. Experiments on challenging SAR data sets demonstrate the effectiveness and potential of the proposed model compared with the current state-of-the-art algorithms.**

***Index Terms*— Change detection (CD), deep clustering, multiscale fusion, octave convolution, self-attention (SA) mechanism, synthetic aperture radar (SAR) images.**

I. Introduction

HUMAN activities have resulted in the fast changes of the Earth's surface. Earth observation by collecting remote sensing satellites data provides an important indicator of interpreting the changes influenced by human beings on the living planet. Change detection (CD) is the process of identifying the changes occurred in an identical geographical area by observing it at different dates [1]. The dynamic phenomenon observation of the Earth's surface with timely and precise CD techniques extremely contributes to understanding connections and interactions between human and natural environment so as to facilitate human decision. Therefore, it is of great importance in researching the CD techniques [2]. The increasing availability of synthetic aperture radar (SAR) images offers unique utility in environmental monitoring purposes [3]–[6].

As we all known, SAR sensor is an active microwave sensor, whose imaging system actively emits the electromagnetic pulses to the Earth's surface and receives the backscattered echoes from the ground objects. By processing and interpreting the received echo signals, the SAR image is formed [4], [5]. The image encompasses the useful information of the terrestrial terrain and can be used to measure the changes for an object at two or multiple observation dates in the same geographical location. Contributed by the microwave imaging principle of SAR image, its imaging process is not affected by the status of sunlight, cloud cover, atmosphere, and so on. Studying environment transition, land use/cover monitoring, and crop management and particularly evaluating damages when a natural disaster happens, SAR images are a significant information resource for CD. However, due to the coherent disturbance of SAR sensor from target scatters, a peculiar effect can be observed in SAR images, namely speckle [7]. It is of the nature of multiplicative noises, being mixed up with the changes of interest. When a CD project is applied, the speckle noises often trigger the misdetection and highly false alarm rates. The detection performance is considerably decreased. Hence, how to reduce the effect of speckle noises is the main challenge in the domain of SAR image CD [8].

Up to now, many researchers have proposed multitudinous methods to better identify the Earth's surface changes [9], [10]. These methods can be broadly grouped into supervised methods and unsupervised methods [11], [12]. Supervised methods need available ground-truth information in its learning process [13]. The common method is converting CD problem into a classification task, where changed and unchanged pixels

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are inferred by the learned classifier, such as support vector machine and extreme learning machine (ELM) [14], [15]. By contrast, unsupervised methods have broader needs than the supervised ones in many practical applications, such as disaster assessment related to fire, flood, and Earthquake, because it is so difficult to obtain the ground-truth labels for these remote sensing data in advance.

Hence, unsupervised CD methods have been extensively studied and applied from a practical perspective. As many studies demonstrated, they usually associate with differencing operators. In terms of simplicity and effectiveness, log- and mean-ratio operators are common methods to obtain a difference image (DI) in two temporal images. Compared with difference operator, ratio operators can better resist speckle noises since it can transform multiplicative speckle into additive ones. Based on ratio operator, other improved methods are proposed, such as neighborhood-based ratio approach [16], generalized likelihood ratio test-based method [17], iterative feedback-based method [18], wavelet fusion method [19], and Gauss-log ratio image fusion-based method [20]. These DI generation algorithms aim to highlight changed information and suppress unchanged background information so that they enable the subsequent operations to easily divide them into changed and unchanged classes. The subsequent operations mainly include threshold methods and clustering methods. In the context of threshold methods, the statistic distribution of data needs to be described by possible parametric models. For example, the Gaussian model is considered in the Kittler-Illingworth (KI) threshold selection algorithm. To better fit the distributions of changed and unchanged classes in a log-ratio image, Bazi *et al.* [3] reformulated the KI criterion under the generalized Gaussian assumption. Without modeling any data distribution, the clustering method can be used more flexibly than the threshold one. It has been reported wide extensions or combinations with other advanced techniques, such as Markov random field [21], principal component analysis [10], kernel method [22], saliency extraction [23], and deep neural networks (DNNs) [24]. The corresponding studies have proved their ability to improve the performance of clustering. However, most of them only explore shallow or handcrafted features to detect the changes, which shows the limited ability to represent the key information hidden in given data.

In recent years, DNNs have obtained a wide and prominent success in the field of image processing, such as image classification, semantic segmentation, and remote sensing interpretation [25]–[29]. Due to the powerful abilities of deep network in dealing with nonlinear transformation, extracting deep features, and flexibly designing hierarchical structures according to practical demands, some researchers also have successfully applied deep models to the task of SAR image CD [30], [31]. Most deep learning-based CD methods take two stages to perform CD for SAR images: 1) predetection—traditional CD methods such as clustering methods are first used to obtain a certain of labeled data and 2) training a classifier—based on the labeled data, a deep model is trained to extract abstract and representative features and then distinguish these features into changed and unchanged classes with a softmax or sigmoid layer [20]. This two-stage pipeline can be as unsupervised methods since it does not use ground-truth information. Gong *et al.* [32] first proposed this pipeline for SAR image CD based on deep belief networks (DBNs), where pixel pairs within a fixed window are cascaded to extract the discriminative features for classifying. Similarly, Gao *et al.* [33] explored a PCANet model to extract representative neighborhood features and directly obtained the change map by a trained classifier. Gao *et al.* [34] recently introduced convolutional-wavelet neural networks (CWNN) for sea ice CD in SAR images, where pooling layers are replaced by dual-tree complex wavelet transform for maintaining more structural information of data. By flexibly designing predetection methods and network architectures, this two-stage pipeline is also popular for CD in other types of data, such as [35]–[37]. Liu *et al.* [37] proposed a local restricted convolutional neural network (CNN) for polarimetric SAR image CD. Wang *et al.* [35] proposed a CNN-based method for CD in hyperspectral images, where subpixel information is mined to enhance the discrimination of changed features. However, in these methods, the quality of labeled data obtained by predetection limits the performance of the classifier trained. Thus, a better unsupervised scheme is necessary to explore for alleviating the limitation and achieving a better detection result. Besides, these methods only consider a local neighborhood with fixed window for pixelwise classification, which cannot fully model and preserve the geometrical characteristics of changed targets. In terms of the local property of vanilla convolution, global difference features do not be considered either. More importantly, due to the complexity of the changed scenes, in all, it is still an open problem for more accurate and reliable algorithms capable of performing SAR image CD in a fully automatic manner.

As mentioned earlier, clustering algorithm has been studied extensively in computer vision. As an unsupervised technique, it can automatically group similar features and recognize hidden patterns [10], [24], but the distribution of raw SAR data is not always clustering-friendly, especially because of the presence of speckle noise. In feature extraction, CNN has unique advantages, i.e., can learn high-level hierarchical features, and is superior to other machine learning methods, but it relies heavily on labeled data for training. Hence, combining unsupervised clustering with CNN to learn clustering-friendly feature representations, one can benefit from the complementary of their advantages, which is promising to detect changes more accurately in an unsupervised manner. Based on the analysis above, in this article, we propose an unsupervised difference representation learning model for CD in SAR images, called multiscale self-attention (SA) deep clustering based on octave convolution (MSDC).

First, a clustering method is combined with CNN into an end-to-end learning framework, namely deep clustering. Unlike most of existing deep learning-based CD methods, we do not train our deep convolutional model with common classification pipeline. Instead, we aim to learn clustering-friendly difference features (CDF) that are suitable for clustering, in which CDF is completed by learning a CNN. In our method, the overclustering assignments provide the label prior for CNN, which is reliable to find similar patterns in data.

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Due to the interference of speckle noise, K-means++ (KM++) is utilized to provide great noise resistance in our unsupervised learning framework.

Second, to drive the end-to-end joint optimization to be effective for learning CDF, we use the Siamese network architecture to extract features from bitemporal SAR images. Followed by two feature-extraction subnetworks, a differencing feature layer is designed to measure the difference level between obtained feature pairs and highlight changed areas, making the network easy to capture semantic changed information. Moreover, octave convolution with a small volume of parameters on SAR images is utilized to extract spatial features from high- and low-frequency aspects, and SA mechanism is introduced to capture the interested spatial areas that contain significant information for the difference feature extraction. With this, the receptive field of the network is increased in different ways. Through their combined effect on extracting significant global spatial features, the network is equipped with strong antinoise capability, by which good clustering assignments are obtained by the discriminative difference measurement, in turn, the good clustering assignments contribute to better learning the CDF for the network.

Last but not least, we extend the proposed model into a multiscale framework by introducing a multiscale fusion module, which receives multiscale blocks as input to construct the pixel representation across scales. Multiscale summation and multiscale SA modules are concretely designed to obtain the multiscale complementary information from multiscale input blocks. As a result, the context information around central pixel is increased effectively from different scales, which is a natural way to convey the statistical property of a central pixel more accurately than a single fixed window, so that it enables difference measurement between pixel pairs to be more discriminative and refines the CDF.

The major contributions of this article can be summarized as follows.

1) We propose a difference representation learning framework by combining KM++ clustering and deep convolutional model, which can be jointly optimized without supervision, providing a new access to identifying the changes from the view of unsupervised deep clustering.
2) The joint optimization of unsupervised clustering and CNN is very challenging to handle for SAR image CD. In order to satisfy the requirements and make an effective optimization process, we tailor a specific CNN, in which the SA mechanism and octave convolution are introduced to enhance the clustering efficiency of the produced features on meaningful areas of the difference, by means of capturing long-range dependencies and extracting information at different spatial frequencies at once.
3) To refine the feature extraction while reducing noise, we propose multiscale fusion strategies, where multiscale blocks of each pixel are fed into the module and then be aggregated by summation or SA mechanism. The obtained multiscale complementary representations are subsequently used to perform the discriminative difference measurement, which helps with the edges and fine details and avoids the limitation of fixed window to a certain extent.

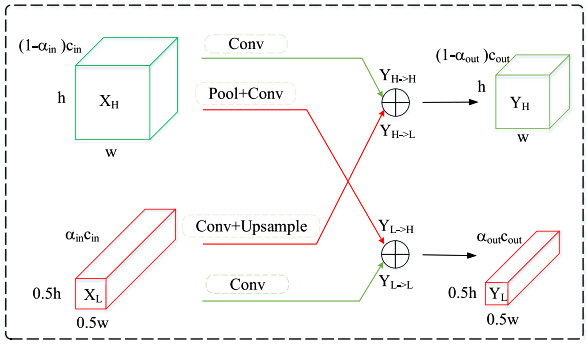


Fig. 1. Octave convolution that decomposes the output feature map from a convolutional layer into a low and a high spatial frequency part. Higher frequency encodes fine details, and lower frequency captures global structural information and reduces spatial redundancy.

The remainder of this article is organized as follows. Section II provides the related work. Section III presents the proposed method. Experimental results are illustrated and compared in Section IV. Finally, Section V provides concluding remarks.

II. Related Work

A. Clustering With Deep Learning

Very recently, deep learning-based clustering has shown advances in unsupervised learning [38]. It jointly optimizes for clustering and DNNs. With the ability of nonlinear mappings, DNN provides the possibility that more clustering-friendly representations are transformed from data without manual feature design. Some works have proved that it is a good idea to joint deep learning and clustering [24], [39], showing that some amount of overclustering is distinctly beneficial as pseudo-labels to extract discriminative representations. However, very few works have tried to combine them for CD, such as [39], where a stack of fully connected layers is used to learn Gaussian-distributed and discriminative difference representations. The used 1-D network provides limited capabilities to extract discriminative features since it does not keep the spatial structure of data. Besides, overclustering labels are only used to pretrain the network, and clustering losses have been jointed to learn the network features and clustering. In this article, we consider the end-to-end learning of CNN features that can work with any clustering algorithms in a more flexible manner, and the well-designed structure of CNN can learn more important spatial information that is beneficial to detect changes.

B. Self-Attention Mechanism and Octave Convolution

SA mechanism is proposed to learn long-range dependencies, which can overcome the disadvantages of vanilla CNN, i.e., only learn local correlation structures within a particular layer. A typical work is nonlocal neural networks [40], which generalizes the classical nonlocal mean operation to DNNs for video classification. Zhang *et al.* [41] applied the SA mechanism to generative adversarial networks and improved

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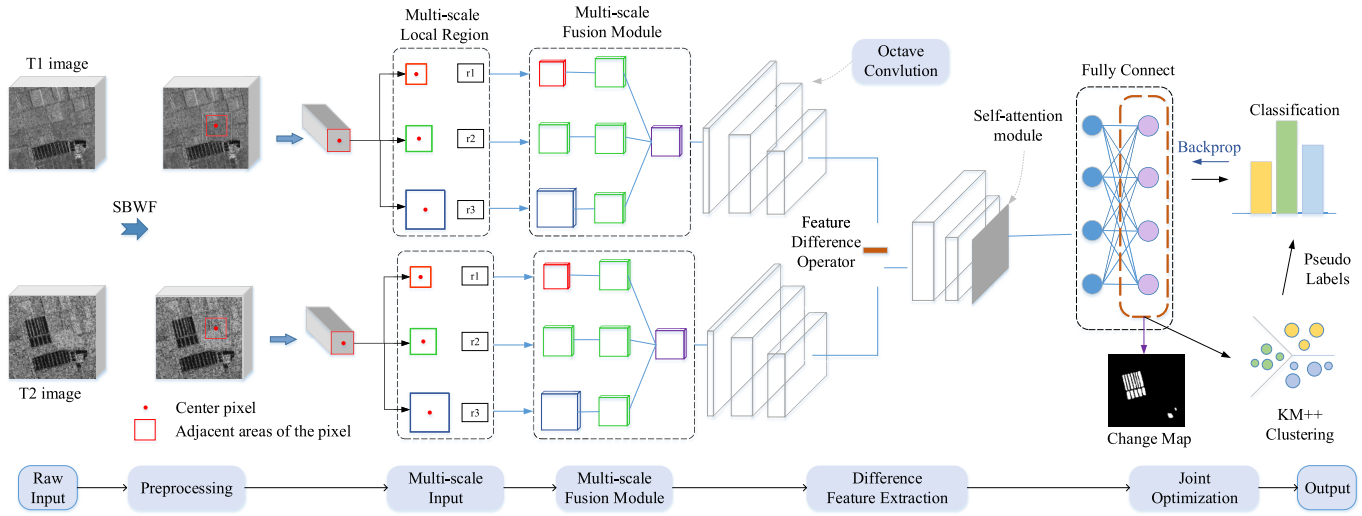


Fig. 2. Overall flowchart of the proposed MSDC.

the quality of generated images. Taken of different route, Chen *et al.* [42] factorized the output feature maps of CNN into a low and a high spatial frequency ingredient shown in Fig. 1, namely octave convolution.

The vanilla convolution takes the feature maps of input and output with the same spatial resolution. In octave convolution, the input feature maps $X \in \mathbb{R}^{c\times h\times w}$ along the channel dimension are decomposed into $X = \{X_H, X_L\}$, where the high-frequency part $X_H \in \mathbb{R}^{(1-\alpha_{\text{in}})c_{\text{in}}\times h\times w}$ captures fine details and the low-frequency part $X_L \in \mathbb{R}^{\alpha_{\text{in}}c_{\text{in}}\times h\times w}$ encodes global structures. Octave convolution directly operates on the multi-frequency representation X by four branches that consist of different combinations of convolution (Conv), pooling (Pool), and upsample (Upsample). Then, the intrafrequency information $Y_{H->H}$ and $Y_{L->L}$ is updated and the interfrequency one $Y_{H->L}$ and $Y_{L->H}$ is exchanged. We can obtain the output feature maps with high and low frequencies as $Y = \{Y_H, Y_L\}$, where $Y_H = Y_{H->H} + Y_{L->H}$ and $Y_L = Y_{L->L} + Y_{H->L}$. For the first and last octave convolution of the network, one sets $\alpha_{\text{in}} = 0$ and $\alpha_{\text{out}} = \alpha$ and $\alpha_{\text{in}} = \alpha$ and $\alpha_{\text{out}} = 0$, respectively, and the middle octave convolution of the network is set as $\alpha_{\text{in}} = \alpha_{\text{out}} = \alpha$, where $\alpha \in [0, 1]$ represents the ratio of channels of the low-frequency part to the overall number of channels. Since the low-frequency maps encode the global structures with a low-resolution tensor, the benefits of octave convolution are threefold: reducing spatial redundancy, increasing receptive field, and thus certain immunity to noise.

In this article, to keep the advantages of deep clustering framework for CD in SAR images, we exploit an effective CNN architecture by integrating SA mechanism and octave convolution into the network, which greatly contributes to the success of deep clustering to identify for changes and unchanges while avoiding the inferences of complex background and speckle noise in SAR images. The main idea is to increase the noise resistance of the network by enlarging its receptive field to extract important global spatial features.

Besides, to make full use of the ability to capture global features via SA mechanism, we propose multiscale SA modules to guide the fine fusion across scales, ensuring the perfectibility of difference measurement.

III. Methodology

A. Proposed MSDC Model

Let us consider two SAR images X^{T1} and X^{T2}, which are acquired at the same geographical area over the different times T1 and T2. The two SAR images suffer from the contamination of speckle noises, and the basic CD methods usually follow three steps to obtain the changes: 1) image preprocessing; 2) DI generating; and 3) DI analyzing. To better detect the changes, image denosing is often applied in preprocessing. As shown in Fig. 2, the proposed model integrates these steps into a unified framework for SAR image CD. When optimization, it alternates between the clustering and updating the parameters of CNN.

In this framework, the features obtained by the well-designed CNN are clustered first, and the generated cluster assignments are used as supervisory information to train the network for learning the CDF. Therefore, the proposed model is fully unsupervised since no ground truth is used. This is crucial for the field of remote sensing that highly lacks labeled training sets. Then, benefiting from the powerful feature extraction via CNN, the difference features gradually become cluster-friendly as the network is trained and are superior to the original image features and manual design features. From bottom to top, our unsupervised model is separated into five main blocks, which are described one by one as follows.

1) Preprocessing: To acquire a discriminative feature representation for CD, a denoising algorithm is first employed to alleviate the effect of the contamination caused by speckle. Usually, the major preprocessing techniques include geometrical registration, radiometric rectification, atmospheric correction, and denoising methods, in which the first three are indispensable before detection. They enable two SAR images to be comparable from spatial domain and spectral domain, respectively. As recently investigated by Wang *et al.* [43],

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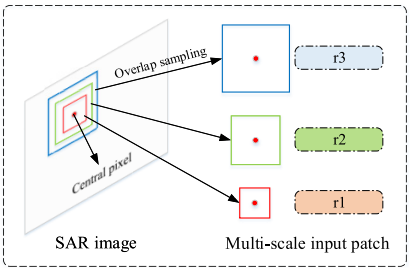


Fig. 3. Multiscale sampling strategy, where $r1$, $r2$, and $r3$ are three different scales for neighboring window around a central pixel, respectively.

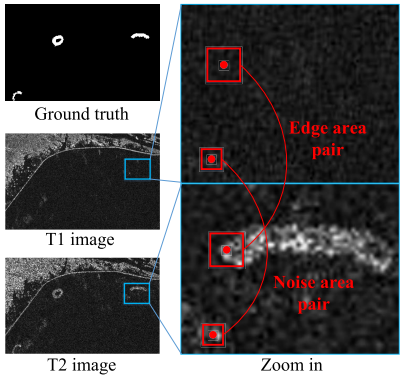


Fig. 4. Illustration of pixels in edge and noise area pairs processed by a single square-shaped window.

the authors argue that structure-preserving despeckling algorithms will benefit the suppression of the speckle and the enhancement of information obtained from changed parts. However, the traditional denoising methods, such as median filtering [44], mean filtering [45], and box filtering [46], and they have restrictive ability to keep the edge information of images while denosing. Therefore, side window filtering (SWF) is employed as the despeckling method for preprocessing in this article since it can not only reduce noises but also preserve good edge information that enhances structure-preserving [47]. For an instance of box filtering, side window box filtering (SWBF) is specifically used in our method. The SWBF technique changes the square window used in box filtering to the side window, which is surprisingly simple yet very effective to retain the edges in practice. This will benefit our method to suppress certain speckle noise yet not lead to the loss of useful information for obtaining meaningful changes.

2) Multiscale Input: After SWBF, we sample multiscale adjacent regions as a basic difference analysis unit for a given central pixel (see Fig. 3). The central pixel and its neighbor often comprise similar materials and have a high probability of belonging to the same class. However, existing CD algorithms usually adopt the fixed window of single scale to sample training data (e.g., 3×3 and 5×5), which cannot universally adapt to data sets with diverse target distributions and is difficult to balance against noise removal and the maintenance of changed details.

To better illustrate, Fig. 4 shows the pixels in edge and noise area pairs of real SAR images considered by one square-shaped region. Suppose that one spatial feature pair of big red box extracted for an edge pixel belongs to changed class.

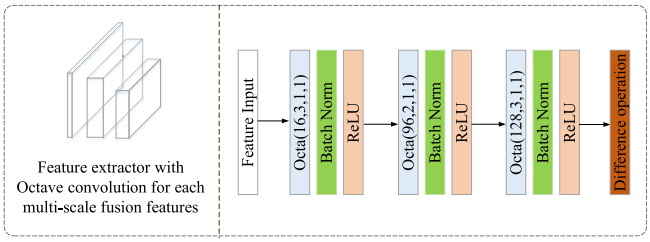


Fig. 5. Detailed framework of feature extractor for each multiscale fusion features f_{out}.

However, most pixels in the extracted area are the same for the patch pair. Obviously, the detection result for the pixel of interest by using this single large window will be detected falsely. For correct detection, a smaller window would be better in this edge pixel pair. Nevertheless, one spatial feature pair of a small red box extracted for a noise pixel is highly possible to be disappointingly detected as changed pixels. Accordingly, the larger neighborhood size is conducive to noise removal but will lead to detail loss, while the smaller window size can better maintain details but easily introduce noise. Hence, by considering the multiscale adjacent spatial regions for each pixel, it is more suitable and powerful to model the geometrical characteristics of various target distributions so that it retains more changed details while reducing noise. We empirically consider the three scale square blocks to jointly extract the spatial features around a central pixel shown in Fig. 3.

3) Multiscale Fusion Module: We adopt the Siamese network architecture that contains two subnetworks to process X^{T1} and X^{T2} in parallel. Each subnetwork receives a sequence of multiscale blocks of the central pixel as input with three branches. Each branch corresponds to a scale of input. The branch that receives the same scale blocks as input shares weights. A multiscale fusion module is designed to capture the multiscale collaborative representation from the input across scales. Through this module, not only multiscale fusion information can be obtained, but also the complexity of the model can be reduced effectively. Specifically, the network with the two parallel subnetworks, i.e., six branches, is pruned into two branches by the multiscale fusion module. Hence, our network becomes more lightweight than the one with six branches and, meanwhile, explores the multiscale feature maps with more semantic and spatial information for the subsequent difference measurement across scales, making it more discriminative and refined. As shown in Fig. 2, we will elaborate the multiscale fusion module in Section III-B. The fused multiscale features are fed into the following convolutional network with Siamese architecture to extract the higher level multiscale fused features.

4) Difference Feature Extraction: Let x_{r1}^{T1}, x_{r2}^{T1}, and x_{r3}^{T1} and x_{r1}^{T2}, x_{r2}^{T2}, and x_{r3}^{T2} be the multiscale regions taken from the same coordinate position in two images X^{T1} and X^{T2}. After the multiscale fusion module is applied, the multiscale fused features for two images are generated, denoted by f_{out}^{T1} and f_{out}^{T2}. They are fed into the fused feature extractor with octave convolution, producing the high- and low-frequency features $f^{T1} = [f_H^{T1}; f_L^{T1}]$ and $f^{T2} = [f_H^{T2}; f_L^{T2}]$, respectively. Fig. 5 shows the details of the feature extractor.

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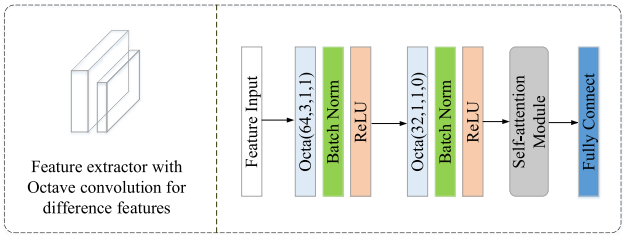


Fig. 6. Detailed framework of feature extractor for difference features f_d.

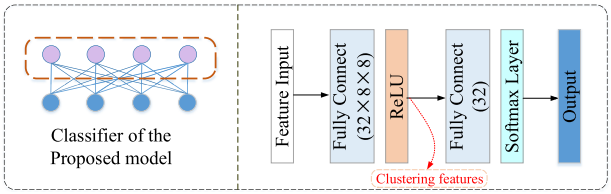


Fig. 7. Detailed framework of the fully connected network.

For each extractor corresponding one image, the parameters are shared. Each convolution process includes convolution, batch normalization (BN), and nonlinear transformation layer in our model. In order to enable the final clustering-features to be differencing-oriented, the difference level between the two multiscale fused features is measured as

$$f_d = \begin{cases} f_{d,H} = \left\| \left(f_H^{T1}\right)^2 - \left(f_H^{T2}\right)^2 \right\|^2 \\ f_{d,L} = \left\| \left(f_L^{T1}\right)^2 - \left(f_L^{T2}\right)^2 \right\|^2 \end{cases} \tag{1}$$

where $\|\cdot\|$ denotes the Euclidean distance. Unlike the common methods that achieve the difference operator in the raw image domain, we apply the difference operator in the domain of multiscale fused high-level features, leading to the refined CDF. In the octave convolution, the feature maps of the high- and low-frequency parts perform the differencing operator, respectively, and generate the high- and low-frequency difference features $f_{d,H}$ and $f_{d,L}$. For simplicity, we denote them by f_d. By this operation, the changed information can be greatly highlighted.

After that, the difference features f_d are further fed into the difference feature extractor with octave convolution. In its ends, the original SA module [40], [41] is added to capture the global difference evidence. We illustrate this difference feature extractor with more implementation details in Fig. 6. Besides, Octa($\cdot$, $\cdot$, $\cdot$, $\cdot$) in Figs. 5 and 6 denote Octa(channels of output, sizes of kernel, stride, padding).

5) Joint Optimization: As the common CNN model, the chain of the difference feature extractor is ended by the fully connected layer shown in Fig. 2. The fully connected network with more implementation details is shown in Fig. 7. Note that the output of the previous layer of the top layer is used to cluster and the clustered result as pseudo-labels provides the supervisory signals for the proposed CNN model. The more details of joint optimization are described in Section III-C. Since the network training is induced by the clustering result, its quality plays a crucial role in our unified framework. As one of the most widely used unsupervised technologies, K-means (KM) clustering is of simplicity and efficiency. However, the original KM algorithm randomly initializes the cluster centers, which leads to the clustering results with local optima and is extremely sensitive to speckle noise. KM++ presented by Arthur and Vassilvitskii [48] is more robust to noises than KM. The authors improve the selection of the initial clustering center based on KM. The key idea is that a point far from the existing cluster center has a greater probability of being selected as the next cluster center. Thereby, KM++ can optimize the quality of the cluster analysis over KM and with high efficiency. Therefore, KM++ is used as an alternative of KM for a better solution.

It is worth noting that only despeckling in preprocessing and introducing SA mechanism are not adequate. Neither the basic KM algorithm nor vanilla convolution possesses enough immunity to strong level of speckle noises. In the learning process, noise is constantly accumulated as iteration increases. To solve this problem, octave convolution and KM++ are further deployed for despeckling. In our network architecture, aside from one original convolutional layer in a multiscale fusion module, the rest of convolutional layers are replaced by octave convolution. In octave convolution, we set $\alpha = 0.75$ throughout the network. The reason is that speckle noise is mostly high-frequency components, and thus, the less high frequency is, the greater noise immunity the network has.

B. Multiscale Fusion Module

As aforementioned, we design our model to receive multiscale neighboring information as input for each pixel shown in Fig. 3. The advantage of multiscale input is that it is a natural way to enhance the context information centered a pixel and reflect the essential characteristics of the center pixel better so that it alleviates the limitation of the fixed window of single scale. In order to maintain the benefits of multiscale input while minimizing the complexity of the model it introduces, we adopt an early fusion scheme by applying designed multiscale fusion modules. By this module, multiscale input information is merged into a complementary feature representation across scales, and not only spatial context information can be enhanced from different scales for each pixel but also the complexity of the model can be reduced effectively. More importantly, the integrated multiscale feature maps with more contextual and semantic information are subsequently used to measure the difference level between pixel pair and enable the generated difference representation to be with strong separability and fine details.

For each image X, a multiscale module is utilized to extract the multiscale fusion information. To let the model adaptively find the best fusion weights on scales, two kinds of multiscale fusion strategies are designed, namely, multiscale summation (without SA) and multiscale SA (with SA) modules. The module detail is shown in Fig. 8, in which Conv($\cdot$, $\cdot$, $\cdot$, $\cdot$) denotes Conv(channels of output, sizes of kernel, stride, padding). According to adaptivity, they are further divided into four types of multiscale strategy, namely, fixed multiscale (FM), adaptive multiscale (AM), SA-based multiscale (SM), and SA-based adaptive multiscale (SAM). Concretely, for different branches of each subnetwork, the different convolution kernels W_q with

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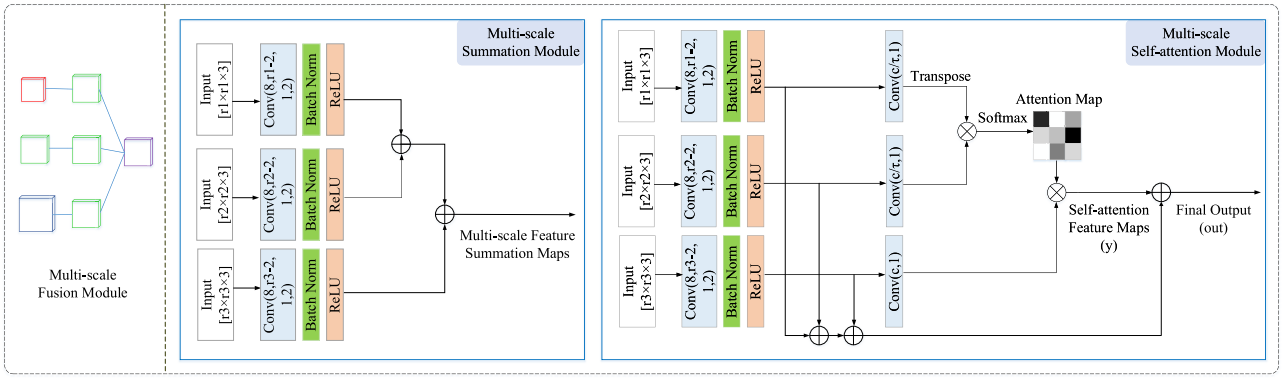


Fig. 8. (Left) Detailed description of the multiscale summation module. (Right) Detailed description of the multiscale SA module. $\oplus$ represents the matrix summation. $\otimes$ represents the matrix multiplication. The softmax operation is performed on each row.

the bias b are employed

$$Z = W_q * x_q + b, \quad q \in \{r1, r2, r3\} \tag{2}$$

where $*$ denotes the convolution computation, Z is the output, and W_q denotes the convolutional kernel with the size $q-2$, which can align the different scale input to produce the same scale output so that it trains a CNN model uniformly without any scale obstacles. Following it, BN [49] is adopted to normalize each output $Z_b = W * x$ in a batch, making its distribution around the mean of 0 and the variance of 1. Consider a mini-batch $Z' = \{Z'_1, Z'_2, \ldots, Z'_m\}$ sampled from the whole sample sets and of the size of m. The normalization is computed as

$$\hat{Z} = \gamma \frac{Z' - E(Z')}{\sqrt{\mathrm{Var}(Z') + \epsilon}} + \beta \tag{3}$$

where $\hat{Z} = \{\hat{Z}_1, \hat{Z}_2, \ldots, \hat{Z}_m\}$ is the normalized output in the mini-batch Z', $E(Z')$ and $\mathrm{Var}(Z')$ denote the expectation and the variance of Z', respectively, and γ and β are the scale and shift parameters, respectively, which are learned along with the parameters of the network. Note that the computation of BN ignores the bias b because its effect can be offset via the above expectation subtraction and β additionally plays the role of the bias. For the nonlinear transformation layer, the output activation value $\widetilde{Z}$ is computed by the rectified linear units (ReLU) [50]

$$\widetilde{Z} = \max\{\hat{Z}, 0\}. \tag{4}$$

After that, we define the above feature-extraction operation for input from one image as

$$f_{\tilde{Z}}^{x_q} = \Phi(x_q, \theta), \quad q \in \{r1, r2, r3\} \tag{5}$$

where Φ represents the convolution process consisting of convolution (conv), BN, and nonlinear transformation, x_q denotes the input of specific scale, $f_{\tilde{Z}}^{x_q} \in \mathbb{R}^{c \times h \times w}$ is the features extracted from x_q, and θ represents the network parameters including W, b, γ, and β. For the subsequent convenience, $f_{\tilde{Z}}^{x_q}$ is denoted by f^{x_q}.

Then, the feature maps f^{x_q} with the same size are obtained from the multiscale input.

1) FM: FM module is the direct summation of the feature maps f^{x_q} as

$$f_{\mathrm{out}} = f^{x_{r1}} + f^{x_{r2}} + f^{x_{r3}} \tag{6}$$

where f_{out} denotes the multiscale fused features.

2) AM: Based on FM, AM is equipped with adaptivity by adding the learnable parameters, λ_1, λ_2, and λ_3, as

$$f_{\mathrm{out}} = \lambda_1 f^{x_{r1}} + \lambda_2 f^{x_{r2}} + \lambda_3 f^{x_{r3}} \tag{7}$$

where λ_1, λ_2, and λ_3 are initially set to 1.

3) SM: SM is based on the SA mechanism. Different from original SA mechanism operating in the identical feature map, we extend it to the different feature maps with the same size obtained by different convolutional kernels from multiscale input. Having such feature maps $f^{x_q} \in \mathbb{R}^{c \times N}$, where $N = h \times w$, since the smaller size is more strongly correlated with the central pixel, $f^{x_{r1}}$ and $f^{x_{r2}}$ are used to compute the attention map as follows:

$$v_{i,j} = \phi\left(f_i^{x_{r1}}\right)^T \varphi\left(f_i^{x_{r2}}\right) \tag{8}$$

$$\mu_{j,i} = \frac{\mathrm{Exp}(v_{i,j})}{\sum_{i=1}^{N} \mathrm{Exp}(v_{i,j})} \tag{9}$$

$\mu_{j,i}$ is the attention map, c is the number of channels, w and h is the wide and height of the features, respectively. Thus, N is the number of the feature position of feature map obtained from the previous convolution layer Φ. The output of the attention layer at the position j is y_j

$$y_j = \sum_{i=1}^{N} \mu_{j,i} \psi\left(f_i^{x_{r3}}\right) \tag{10}$$

so its whole output can be expressed as $y = (y_1, y_2, \ldots, y_j, \ldots, y_N)$. In addition, $\phi(f_i^{x_{r1}}) = W_\phi f_i^{x_{r1}}$, $\varphi(f_i^{x_{r2}}) = W_\varphi f_i^{x_{r2}}$, and $\psi(f_i^{x_{r3}}) = W_\psi f_i^{x_{r3}}$ in the above equations, where $W_\phi \in \mathbb{R}^{\bar{c} \times c}$, $W_\varphi \in \mathbb{R}^{\bar{c} \times c}$, and $W_\psi \in \mathbb{R}^{c \times c}$ are the learnable parameters and are implemented using the convolution of kernel 1×1, and $\bar{c} = c/\tau$, $\tau = \{1, 2, 4, 8, \ldots\}$ is the hyperparameter. In our experiments, τ is set to 2.

Finally, the output of attention layer is fused with the previous multiscale feature maps, obtaining the final output by

$$\mathrm{out}_i = \lambda y_i + f_i^{x_{r1}} + f_i^{x_{r2}} + f_i^{x_{r3}} \tag{11}$$

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where λ is a parameter to be learned and is initially set to 0. This permits that the network first depends on the local features of easy learning, and then, more weight is allocated to the nonlocal features with network training. Intuitively, in contrast to nonlocal dependencies, the local cues are easier to learn. The easy task should be first learned, and then, the complexity of the task is increased progressively.

4) SAM: The difference between SAM and SM is only that the computation of the final fusion output out_i is different. In SAM, it is computed as

$$\text{out}_i = \lambda y_i + \lambda_1 f_i^{x_{r1}} + \lambda_2 f_i^{x_{r2}} + \lambda_3 f_i^{x_{r3}} \tag{12}$$

where λ, λ_1, λ_2, and λ_3 are the learnable parameters along with the network training. λ is initially set to 0, and λ_1, λ_2, and λ_3 are initially set to 1.

The multiscale SA modules (SM and SAM) above are proposed to merge the multiscale input information, in which insight is simple. Since the SA mechanism can capture the long-range spatial dependencies and adaptively learn important global information, it would be able to capture the multiscale spatial dependencies according to its computation rule that considers the global information. The advantages of doing this are twofold: 1) it makes up the weakness of convolution with local receptive field and can integrate global and local features effectively and 2) it serves as pruning and can enable our CNN architecture to be lightweight. Comparatively, summation modules (FM and AM) only equip the ability to integrate local features since it does not use SA mechanism for global feature extraction. For simplicity, we represent the proposed network model with different multiscale fusion strategies as MSDC-FM, MSDC-AM, MSDC-SM, and MSDC-SAM.

C. Optimization Procedure

For convenience, we let $x_l = [x_{q,l}^{T1}, x_{q,l}^{T2}]$, $\{x_l \in \mathcal{X}\}_{l=1}^n$, where n is the number of training samples. Instead of clustering in the raw data domain, we propose to cluster in the latent feature domain that is transformed with a nonlinear mapping. CNN is a popular choice as our mapping function since it achieves the best performance on learning the latent and discriminative features in complex spatial context information [51]. Therefore, a suitable CNN architecture is designed as described above. We define the whole nonlinear transformation with CNN as $f(x_l, \theta)$ and refer to the vector generated by this mapping from SAR image pair as feature representation. The parameter θ of the network is first initialized, producing the initial features $f(x_l, \theta)$. KM++ is implemented by optimizing the following problem:

$$\min_{M \in \mathbb{R}^{m \times k}} \frac{1}{n} \sum_{l=1}^{n} \min_{s_l \in \{0,1\}^k} \| f(x_l, \theta) - M s_l \|_2^2$$
$$\text{s.t. } {s_l}^T 1_K = 1 \tag{13}$$

where s_l is the cluster assignments of the features $f(x_l, \theta)$ and M denotes the $m \times k$ centroid matrix. In turn, these assignments are used to optimize the network parameters θ with cross-entropy loss function, which is defined as

$$H(s_l, p_l) = -\frac{1}{n} \sum_{l=1}^{n} p_l \log s_l \tag{14}$$

Algorithm 1 Optimization Procedure of the Proposed MSDC Model

Input: n pairs of multiscale inputs $x_l = [x_{q,l}^{T1}, x_{q,l}^{T2}]$, where $q \in \{r1, r2, r3\}$ and $l = \{1, 2, \ldots, n\}$.
Output: Change map.
1: Initialize the network parameters θ and produce the initial features $f(x_l, \theta)$.
2: **for** $t = 0 : T$ **do**. (Perform $T + 1$ epochs).
3: **if** $t\%R = 0$:
4: Update cluster assignments s_l by Eq. 13.
5: **end if**
6: Update network parameters θ by Eq. 14.
7: **end for**
8: Cluster the final features into two classes, apply mean filtering and obtain the final CD result.
End

where H quantifies the errors of the predicted labels p_l by comparing to the pseudo-labels s_l. Note that we do not use the centroid matrix of cluster method in the loss function of the network. Overall, the optimization procedure of the whole model is summarized in Algorithm 1. It alternates between two steps: 1) KM++ clusters the features obtained by CNN, generating the cluster assignments, and 2) the cluster assignments are used as pseudo-labels to update the network parameters, enabling better features to be mapped by the network. In Algorithm 1, R is the parameter that is set to control the interval of cluster reassignment, i.e., update the network parameters R times after clustering the features once.

IV. Experiments and Analysis

A. Experimental Data and Evaluation Criterion

In order to evaluate the effectiveness of the proposed method, seven SAR image pairs are used in our experiments shown in Figs. 9 and 10. The first two data sets are taken from the Yellow River data set, which was acquired by Radarsat-2 satellite in June 2008 and June 2009 at the region of Yellow River Estuary in China. It is worth noting that the two images are single-look image and four-look image, respectively. This means that the influence of speckle noise on the former is much greater than that of the latter, which easily causes high false alarms and missed detections, complicating CD. Experiments on such data set with large noise provide a good chance to evaluate the performance of the proposed method. Since the original Yellow River data set is of too large size (7666×7692 pixels), we select the two representative areas with small size to validate the proposed method. They are Farmland and Coastline data sets, corresponding to the sizes of 306×291 and 450×280 pixels, respectively. The third data set is De gaulle airport image pairs, which were taken by ERS-1 SAR sensors in July 1997 and October 1998, with the size of 240×370 pixels. This data set is related to the areas of De Gaulle Airport in France. The fourth data set is Red River image pairs, which were taken from Red River, Vietnam, on August 24, 1996 and August 14, 1999, corresponding to nonflood and flood, respectively. This data set is captured by ERS-2 sensors, with the image size of 512×512 pixels.

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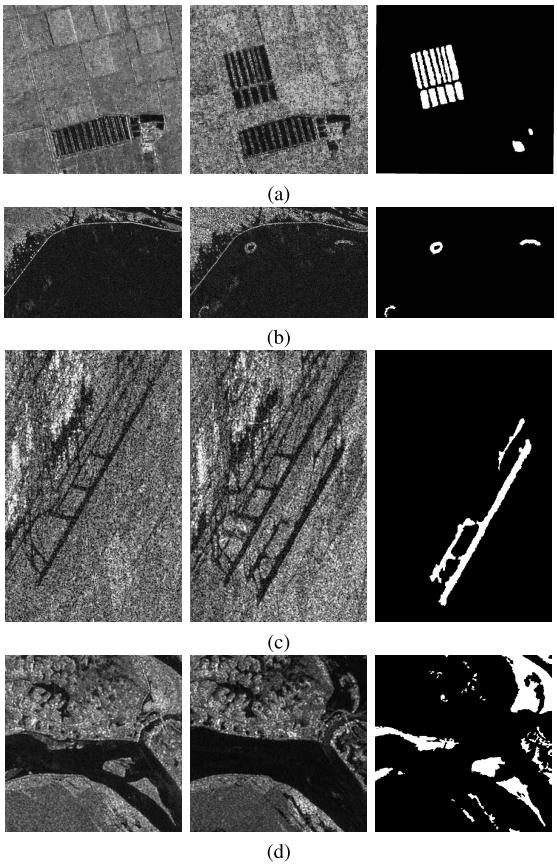


Fig. 9. Test data sets. Four pairs of SAR images and their corresponding ground truth. (a) Farmland data set. (b) Coastline data set. (c) De Gaulle Airport data set. (d) Red River data set.

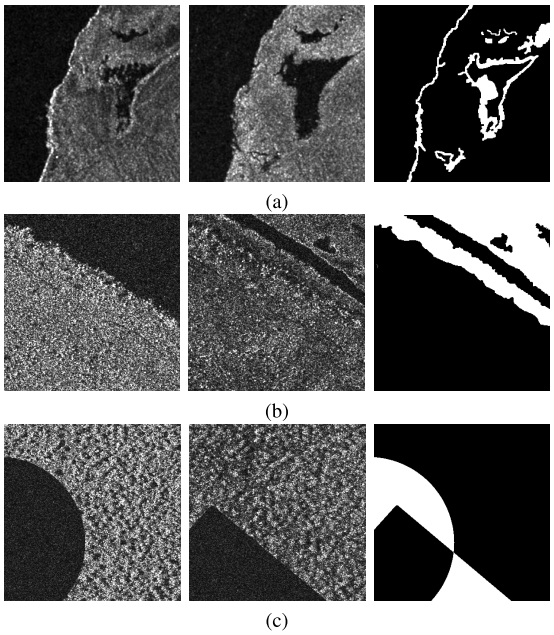


Fig. 10. Test data sets. Three pairs of high-resolution SAR images and their corresponding ground truth. (a) Brazil data set. (b) Namibia data set. (c) Simulation data set.

The last three data sets are high-resolution SAR data, which is used to conduct compared experiments as our attempt at high-resolution SAR CD problem. These three data sets are acquired from the two large high-resolution SAR images of Brazil and Namibia, captured by the TerraSAR-X satellite. The first large image pair (10 878 × 18 958 pixels) is taken in June 2010 and January 2013, respectively. The second one (9376 × 8328 pixels) is taken in April and September 2009. Because the original sizes of both the SAR images are too huge to show the detailed information, we select two typical areas (300 × 300 pixels for each area). The third data are the simulation data set with the size of 400 × 400 pixels, where the two temporal images are cropped from the corresponding temporal unchanged sections of the large Namibia data. The changed sections are artificially constructed by regular shape.

TABLE I
SPATIAL RESOLUTION DESCRIPTIONS OF USED DATA SETS

Datasets	Satellite	Spatial Resolution	Incident Angle
Farmland/Coastline	Radarsat-2	8m	30-50
De Gaulle Airport	ERS-1	30m	23
Red River	ERS-2	10m	23
Brazil	TerraSAR-X	3m	20-55
Namibia/Simulation	TerraSAR-X	2m	20-55

The available ground-truth images of these data sets are generated by manual photointerpretation with prior information. The spatial resolution and incident angle descriptions of these data sets are shown in Table I.

In our experiments, five metrics are used to quantitatively evaluate the performance of the proposed model, i.e., false negative (FN), false positive (FP), overall errors (OE), overall accuracy (OA), and Kappa statistic. FN denotes the number of pixels that are wrongly classified into unchanged areas. FP denotes the number of pixels that are wrongly classified into the changed areas. Then, OE is computed as $\mathrm{OE} = \mathrm{FN} + \mathrm{FP}$. Moreover, true negative (TN) represents the number of pixels that are correctly classified into the unchanged areas, and true positive (TP) represents the number of pixels that are correctly classified into the changed areas. Thereby, OA can be obtained by $\mathrm{OA} = (\mathrm{TP} + \mathrm{TN})/(\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN})$. The kappa coefficient is calculated as

$$\mathrm{Kappa} = \frac{\mathrm{OA} - \mathrm{PRE}}{1 - \mathrm{PRE}} \tag{15}$$

where

$$\mathrm{PRE} = \frac{(\mathrm{TP} + \mathrm{FN}) \cdot (\mathrm{TP} + \mathrm{FP}) + (\mathrm{TN} + \mathrm{FP}) \cdot (\mathrm{TN} + \mathrm{FN})}{(\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN})^2}. \tag{16}$$

A higher kappa value indicates better performance.

B. Experimental Settings

The preprocessing step (SWBF) of the proposed method is run on MATLAB. The other steps are implemented by Python language, and the network is deployed on a single GeForce RTX 2070 SUPER 8-GB GPU and using Pytorch framework with version 1.1.0. During training, the convolutional layers of the network are initialized by drawing weights from zero-mean Gaussian distributions with a standard deviation $(2/(K_i \times K_i \times c_{\mathrm{out},i}))^{1/2}$, where K_i denotes the kernel size and $c_{\mathrm{out},i}$ denotes the number of output channels at the

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TABLE II
DETECTION PERFORMANCE OF DIFFERENT LEARNING RATES FOR SAMSDC ON FARMLAND DATA

	OA (%)	Kappa (%)
0.1	98.72	88.00
0.01	98.95	90.64
0.001	98.64	87.83

ith convolutional layer. The fully connected layers are initialized by drawing weights from zero-mean Gaussian distributions with a standard deviation 0.01. The weights of BN layers and biases are initialized as 1 and 0, respectively. The input data are scaled to the range $[-1, 1]$. Since the network features from random initialization are used to first cluster, the version of Pytorch framework will affect the performance of the proposed algorithm.

The networks are trained with stochastic gradient descent (SGD) optimizer and a momentum of 0.9. Each mini-batch contains 256 sample patches. The features extracted by the proposed network are whitened, and then, the whitened features are clustered into C_n clusters to improve the robustness of clustering. One could reassign the clusters every R epochs as pseudo-labels. After reassigned, sample elements of each class of pseudo-labels are uniformly sampled to the number of $(N_p/C_n) + 1$ as training samples, which is significant for clustering to overcome the problem of sample imbalance, where N_p denotes the number of the total pixels in an input image. After the networks are trained well, the features are finally clustered into two classes, i.e., changes and unchanges. Moreover, in order to reduce the possible isolated points and outliers, the simple median filtering [44] are applied to the binary map, leading to the final change map. Here, we consider the three scale square blocks $(r1, r2, r3) = (3, 5, 7)$ for multi-scale input on test data, which will be studied in Section IV-D.

1) Learning Rate: Different from the weights of the network model that can be learned automatically, the learning rate is an important parameter to be set since it significantly affects the training performance. We vary the learning rate on a logarithmic scale and report the results in Table II. As can be observed, large and small learning rates both degrade the detection performance. The best performance is obtained with learning rate $= 0.01$. Thus, we set the value of learning rate to 0.01 for the proposed model in follow-up experiments.

2) Number C_n of Clusters and Number R of Reassignments Between Epochs: We study the impact of the number C_n of clusters used in KM++ on the quality of the model. The experiments are constructed on the Farmland data set shown in Fig. 11. We have tried the value of C_n with {2, 5, 10, 15, 20}. As can be observed, the performance of the model is the best when $C_n = 15$. This implies that some overclustering is reliable to provide the labeled prior information and learn good representations from given data since overclustering can assign homogeneous regions into the same cluster. At each epoch, we reassign the images to a new set of clusters, with guarantee of stability only when $C_n = 2$. When $C_n = \{5, 10, 15, 20\}$, we assign the images to a set of clusters only at the first epoch, which can guarantee the stability. Therefore, we set $C_n = 15$ and assign the images to a set of clusters only once, i.e., at the first epoch in our experiments.

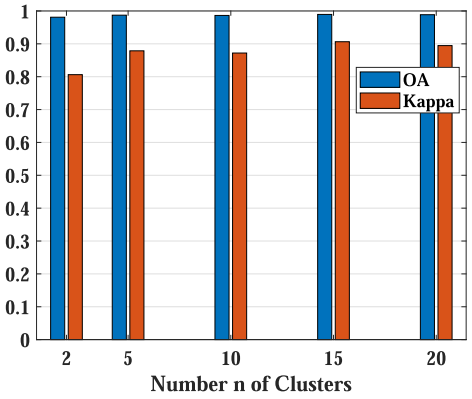


Fig. 11. Effects of the number n of clusters.

TABLE III
QUANTITATIVE COMPARISON OVER DIFFERENT METHODS ON THE FARMLAND DATA SET

	FP(%)	FN(%)	OE(%)	OA(%)	Kappa(%)
PCAKM	1.46	0.55	2.00	98.00	83.20
GaborPCANet	**0.06**	1.46	1.52	98.48	84.66
DBN	0.95	0.56	1.51	98.49	86.81
CWNN	0.53	0.70	1.23	98.77	88.80
MSDC-FM	0.36	0.61	0.97	99.03	91.50
MSDC-AM	0.52	0.45	0.97	99.03	91.25
MSDC-SM	0.50	**0.42**	**0.92**	**99.08**	**91.72**
MSDC-SAM	0.48	0.57	1.05	98.95	90.64

C. Detection Performance

To test the effectiveness of the proposed model, we compare the proposed model with some state-of-the-art CD methods. These methods include PCA-KM [10], GaborPCANet [33], CWNN [34], and DBN [32]. The first three methods are implemented using their publicly available source codes with the default parameters provided in [10], [33] and [34], respectively. The last method is implemented by ourselves in MATLAB language, the parameters provided in [32] are used. We have run five times for all methods, and the best results are reported.

1) Experiments on Medium- and Low-Resolution Data Sets: Fig. 12 shows the CD results generated by different methods on the four pairs of medium- and low-resolution SAR images. For Farmland data set, the evaluation metrics are listed in Table III. As can be observed in the first line of Fig. 12, PCAKM generates some white noisy areas due to its sensitivity to noise. Thus, PCAKM suffers from the high FP value. GaborPCANet suppresses noise well but misses many changed details, and therefore, the FN value of PCAKM is relatively high. Conversely, DBN detects more changed pixels with a low FN value of 0.06%, but unchanged details are not accurately detected. CWNN detects changed details well, but some unchanged areas are wrongly detected as changed one. Compared with them, the proposed method produces higher OA and Kappa values and lower OE one, which illustrates that the proposed model is better on retaining changed details and

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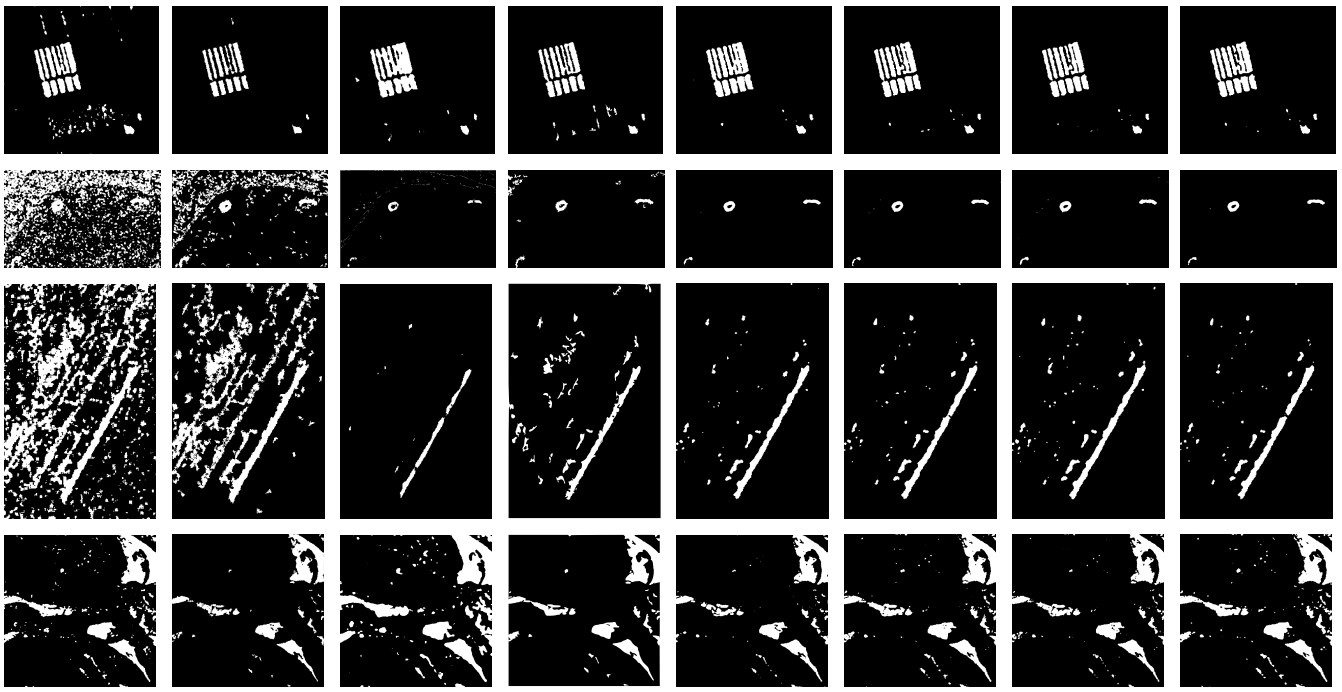


Fig. 12. Experimental results generated by different methods on four pairs of real medium- and low-resolution SAR images. (a) PCAKM. (b) GaborPCANet. (c) DBN. (d) CWNN. (e) MSDC-FM. (f) MSDC-AM. (g) MSDC-SM. (h) MSDC-SAM. The rows from top to bottom represent Farmland, Coastline, De Gaulle Airport, and Red River data sets.

TABLE IV
QUANTITATIVE COMPARISON OVER DIFFERENT METHODS ON THE COASTLINE DATA SET

	FP(%)	FN(%)	OE(%)	OA(%)	Kappa(%)
PCAKM	32.90	**0.01**	32.91	67.09	4.09
GaborPCANet	12.61	**0.01**	12.62	87.38	12.70
DBN	0.40	0.44	0.84	99.16	59.80
CWNN	0.82	0.07	0.90	99.10	68.55
MSDC-FM	0.13	0.08	0.21	99.79	89.82
MSDC-AM	0.13	0.09	0.23	99.77	89.00
MSDC-SM	**0.10**	0.14	0.24	99.76	89.08
MSDC-SAM	0.12	0.08	**0.20**	**99.80**	**90.24**

TABLE V
QUANTITATIVE COMPARISON OVER DIFFERENT METHODS ON THE DE GAULLE AIRPORT DATA SET

	FP(%)	FN(%)	OE(%)	OA(%)	Kappa(%)
PCAKM	24.27	**0.62**	24.89	75.11	16.69
GaborPCANet	13.83	0.66	14.49	85.51	28.62
DBN	**0.07**	2.93	3.00	97.00	46.93
CWNN	2.27	1.64	3.92	96.08	55.71
MSDC-FM	0.70	1.57	**2.27**	**97.72**	**69.58**
MSDC-AM	0.78	1.56	2.34	97.66	69.03
MSDC-SM	0.92	1.51	2.43	97.57	68.63
MSDC-SAM	0.65	1.66	2.31	97.69	68.54

reducing speckle noises than other methods. Besides, the best results are yielded by the proposed MSDC-SM.

For Coastline data, CD maps are shown in the second line of Fig. 12, and the evaluation metrics are reported in Table IV. As we can see, the visual results obtained by PCAKM, GaborPCANet, DBN, and CWNN all contain many white noisy areas with the high FP and OE values, especially for PCAKM and GaborPCANet. Moreover, DBN yields the relatively less noises, but many changed details are not detected, thus suffering from the highest FP value of 0.44%. This indicates that compared methods have a limited ability to suppress speckle noise and retaining changed details. It is evident that the proposed method can generate more accurate results. Among the proposed models, MSDC-SAM produces the highest OA and kappa values of 99.80% and 90.24%, which are far superior to the compared algorithms. Compared with the state-of-the-art CWNN, the kappa value of the proposed MSDC-SAM has increased by 21.99%.

The statistical values and the detection results on De Gaulle Airport data are listed in Table V and exhibited in the third line of Fig. 12, respectively. As shown in Table V, our method outperforms other models, yielding the lowest OE value of 2.27% and the highest OA and Kappa values of 97.72% and 69.58%. The PCAKM, GaborPCANet, and CWNN yield the high FP values of 24.27%, 13.83%, and 2.27%, respectively, which are in line with the visual results. The third line of Fig. 12 shows that their visual results contain many white noisy areas. Moreover, the DBN produces the lowest FP value of 0.07% but misses many changed details. Therefore, the DBN has the highest FN value of 2.93%. Compared with the CWNN, the kappa value of the proposed MSDC-FM has increased by 13.87%. Since these data have a very complex background, the performance of the proposed model on these data indicates its robustness in detecting meaningful changes.

For Red River data, Table VI lists the evaluation metrics, and the fourth line of Fig. 12 shows the visual results.

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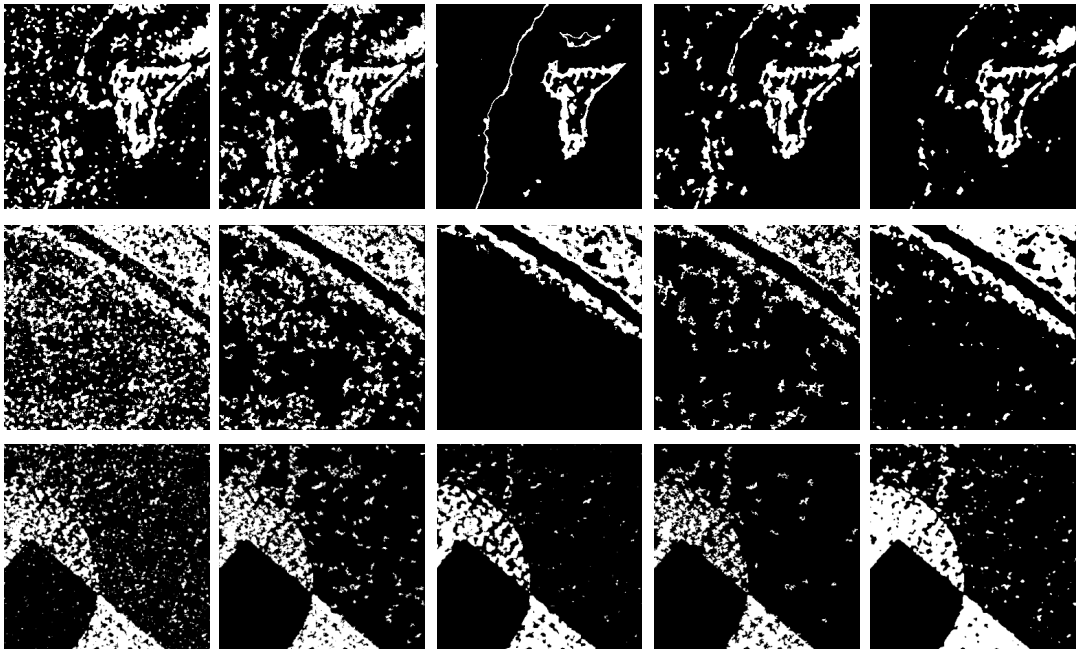


Fig. 13. Experimental results generated by different methods on four pairs of real high-resolution SAR images. (a) PCAKM. (b) GaborPCANet. (c) DBN. (d) CWNN. (e) MSDC-SAM. The rows from top to bottom represent the Brazil, Namibia, and Simulation data sets.

TABLE VI
QUANTITATIVE COMPARISON OVER DIFFERENT METHODS ON THE RED RIVER DATA SET

	FP(%)	FN(%)	OE(%)	OA(%)	Kappa(%)
PCAKM	1.04	1.22	2.26	97.74	89.67
GaborPCANet	**0.06**	4.08	4.14	95.86	78.22
DBN	4.29	0.40	4.69	95.31	81.18
CWNN	0.10	2.81	2.91	97.09	85.43
MSDC-FM	3.30	**0.20**	3.50	96.51	82.25
MSDC-AM	2.19	0.48	2.67	97.33	87.10
MSDC-SM	3.11	0.27	3.37	96.63	83.02
MSDC-SAM	1.40	0.75	**2.16**	**97.84**	**89.98**

TABLE VII
QUANTITATIVE COMPARISON OVER DIFFERENT METHODS ON THE BRAZIL DATA SET

	FP(%)	FN(%)	OE(%)	OA(%)	Kappa(%)
PCAKM	13.42	2.37	15.79	84.21	39.14
GaborPCANet	12.45	**2.32**	14.77	85.23	41.38
DBN	**1.57**	4.87	**6.44**	**93.56**	54.91
CWNN	6.58	3.18	9.77	90.23	50.51
MSDC-SAM	3.13	3.93	7.06	92.94	**56.76**

TABLE VIII
QUANTITATIVE COMPARISON OVER DIFFERENT METHODS ON THE NAMIBIA DATA SET

	FP(%)	FN(%)	OE(%)	OA(%)	Kappa(%)
PCAKM	18.71	9.85	28.56	71.44	30.65
GaborPCANet	12.16	11.25	23.40	76.60	36.87
DBN	**1.60**	11.71	13.31	86.69	57.56
CWNN	5.28	14.43	19.72	80.28	38.11
MSDC-SAM	2.99	**9.77**	**12.76**	**87.24**	**61.45**

From Table VI, we can see that the proposed model with adaptive multiscale, i.e., MSDC-AM and MSDC-SAM, generates better results than ones without adaptive setting, i.e., MSDC-FM and MSDC-SM. Moreover, the model with adaptive SA module, i.e., MSDC-SAM, is superior to the model without SA, i.e., MSDC-AM. This indicates that SAM is of more generalization for different scales of structures than other multiscale strategies in different data. In addition, GaborPCANet and CWNN miss many changed details and thus produce the relatively high FN values of 4.08% and 2.81%, respectively. For the DBN method, the visual result contains some white noisy areas, which lead to the high FP value of 4.29%. In contrast, MSDC-SAM yields the highest OA and kappa values of 97.84% and 89.98%, which slightly outperforms the PCAKM that yields the OA and kappa values of 97.74% and 89.67%. The comparison also shows the effectiveness of the proposed methods on these data.

Based on the comparison analysis above, the proposed model achieves the state-of-the-art performance on the four real medium- and low-resolution SAR data sets. Moreover, the model of MSDC-SAM has the better generalization and robustness in multiple data sets than other multiscale ones because it can adaptively integrate local and global information.

2) Experiments on High-Resolution Data Sets: The detection results generated by the proposed model and the four comparative methods on high-resolution data sets are presented in Fig. 13. Tables VII–IX list the quantitative results. To avoid repetition, we only show the results obtained by the proposed MSDC-SAM since it has better generalization as stated in the above experiment. For fair comparison, the parameters of all algorithms are set as above.

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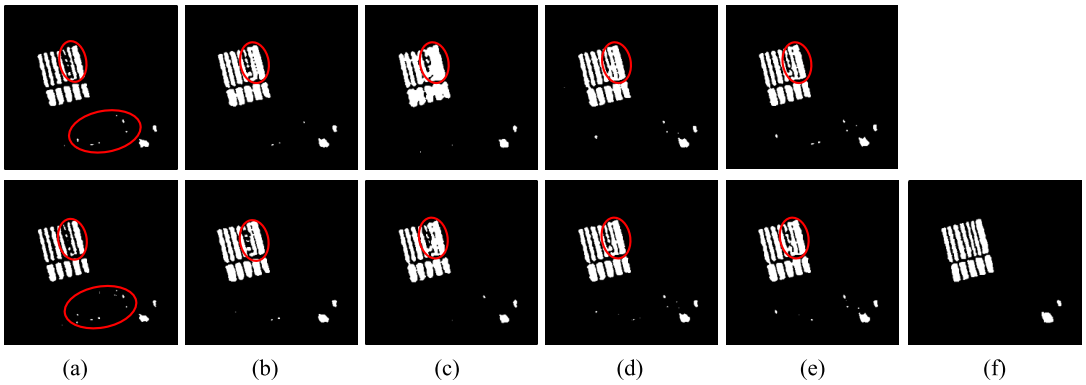


Fig. 14. Comparison results of diverse inputs on Farmland data set. (a)–(c) Results with the single input scale of 3, 5, and 7. (d) and (e) Results of adaptive and nonadaptive multiscale input strategies, respectively. (f) Ground truth. The first and second rows represent without SA and with SA, respectively. (f)

TABLE IX
QUANTITATIVE COMPARISON OVER DIFFERENT METHODS ON THE SIMULATION DATA SET

	FP(%)	FN(%)	OE(%)	OA(%)	Kappa(%)
PCAKM	2.99	9.77	12.76	87.24	61.45
GaborPCANet	4.14	5.95	10.09	89.91	65.15
DBN	**1.07**	6.12	7.19	92.81	73.29
CWNN	1.54	7.86	9.41	90.59	63.96
MSDC-SAM	3.13	**2.55**	**5.68**	**94.32**	**81.35**

Compared with low- and medium-resolution SAR images, high-resolution SAR images are more seriously disturbed by the complexity of targets and imaging noise. Thus, it is more difficult to detect the changed areas for high-resolution SAR images. It can be observed from Fig. 13 and Tables VII–IX, PCAKM and GaborPCANet both generate many white noise areas, and unchanged areas cannot be detected well in the three high-resolution images. Therefore, the FP and FN values of PCAKM and GaborPCANet are relatively high. DBN can reduce the noise well, but many changed details are missed. In each of the three data sets, DBN yields the lowest FP values of 1.57%, 1.60%, and 1.07%, respectively, but also yields the high PN values. CWNN does not perform better than DBN, yielding the higher PN values than DBN in the last two data sets. Moreover, for Brazil data, the proposed algorithm does not yield the best values of OE and OA, but it achieves a better balance between false alarms and missed detection, thus obtaining the best kappa value of 56.76%. Comparatively, in the three data sets, the proposed method improved the kappa value by 1.85%, 3.89%, and 8.06% than the DBN algorithm, respectively. To sum up, under the respective recommended parameters of all methods, the proposed method achieves better performance than compared methods, which greatly declares the availability and superiority of the proposed method on different data sets with diverse target distributions.

D. Analysis on the Multiscale Input

In order to validate the proposed multiscale input strategy described in Section III-A, we study the detection performance using square-shaped regions of different single sizes, i.e., (3, 5, 7). Fig. 14 shows the detection results, and Table X lists the quantitative evaluations, OE, OA, and kappa coefficient, with varying different window sizes. Moreover, we further divide them into two classes to study, i.e., the multiscale fusion module with SA strategy (SA-based multiscale module) and without SA strategy (multiscale summation module).

TABLE X
DETECTION PERFORMANCE OF DIVERSE INPUTS ON THE FARMLAND DATA SET

	Diverse Inputs	OE(%)	OA(%)	Kappa(%)
Without Self-attention	3	1.35	98.65	86.78
	5	1.27	98.73	88.76
	7	1.62	98.38	86.04
	MSDC-FM	0.97	99.03	91.50
	MSDC-AM	0.97	99.03	91.25
With Self-attention	3	1.57	98.43	84.24
	5	1.35	98.65	87.73
	7	1.54	98.46	85.97
	MSDC-SM	0.92	99.08	91.72
	MSDC-SAM	1.05	98.95	90.64

From Table X, we can see that the single block-derived input produces lower accuracy than multiscale input, for which the single scale input has a limited ability to represent surrounding semantic and spatial information at different sizes. As shown in Fig. 14, the smaller single scale window easily introduces noises, and the larger one is more robust to noise, but they both cannot keep changed details well, especially for the areas marked with the red ellipses. Obviously, multiscale input provides a better balance between noise removal and the preservation of changed details. Besides, Table X shows that single block-derived input without SA module outperforms one with SA module, and the best result is obtained by MSDC-SM that uses multiscale inputs with SA mechanism. It is reasonable to infer that the SA mechanism captures the multiscale spatial dependencies as we analyzed in Section III-B. Furthermore, for the single scale input, it is in line with the original introduction by Wang *et al.* [40], i.e., the nonlocal mechanism can capture long-range dependencies, showing that adding the attention mechanism at a relatively deeper layer tends to be better in performance. Because of that, we add an original SA module to the end of feature extraction shown

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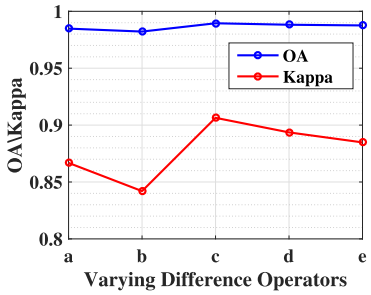


Fig. 15. Effects of varying difference operators: a: $||a| - |b||$, b:$a^2 - b^2$, c: $\|a^2 - b^2\|^2$, d: $|a^2 - b^2|$, and e: $\||a| - |b|\|^2$, where $a = f^{T1}$, $b = f^{T2}$, and $\|\cdot\|$ and $|\cdot|$ denote the Euclidean distance and absolute value, respectively.

TABLE XI
DETECTION PERFORMANCE OF VARYING DIFFERENCE LAYERS ON THE FARMLAND DATA SET

Difference Layer	OA (%)	Kappa (%)
2	98.64	88.36
3	98.95	90.64
4	97.91	80.81

in Fig. 2, whose influence on the detection performance is comparatively analyzed in Section IV-F. Besides, we also try the multiscale input (5, 7, 9), (7, 9, 11), and so on, which do not perform better than (3, 5, 7).

E. Analysis on the Varying Difference Operators and Difference Layers

To evaluate the influence of difference operator on the final accuracy, we plot the curves of detection performance of diverse difference operators on the Farmland data set. As shown in Fig. 15, these difference computations are designed based on the commonly used subtraction operator. This figure shows that although they are slightly different in computation, the detection accuracies present the large difference on this data set. Among them, the used difference operator c is better than other variants, generating the highest OA and Kappa values. This indicates that the difference operator c has a better ability to resist noise and retaining changed details in the proposed model for SAR image CD.

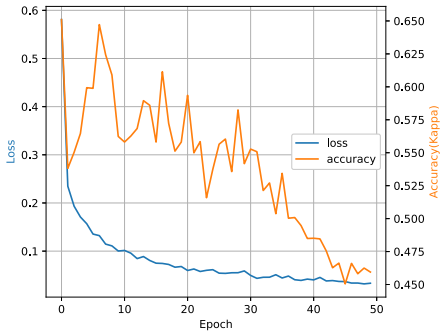
Moreover, we have also studied the effect of varying difference layers on the performance of the proposed MSDC-SAM. Our network architecture consists of 11 layers, in which five layers are octave convolution for feature extraction. For clarity, we take the first layer of octave convolution as the first layer and vary the difference layer from 2 to 4 in order. Table XI lists the quantitative evaluations with varying difference layers. The best result is yielded by setting the difference layer to the third layer of octave convolution. This indicates that it can obtain better performance for the difference layer that is set close to the middlemost layer of the proposed model.

F. Analysis on Without Preprocessing, Vanilla Convolution, KM, and Without the Last SA

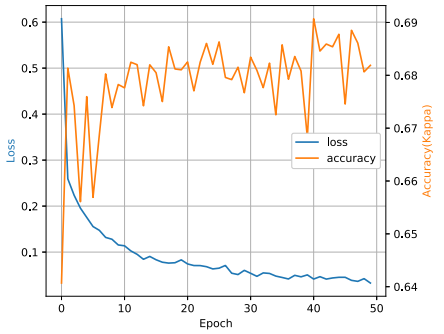
Table XII reports the accuracies of kappa values testing on some alternatives, i.e., without preprocessing, vanilla convolution, and KM. From Table XII, we can see that any setting

TABLE XII
DETECTION PERFORMANCE OF SOME ALTERNATIVES OVER WITHOUT PREPROCESSING, VANILLA CONVOLUTION, AND KM

Without Preprocessing	Vanilla Convolution	KM	Kappa(%)
√			88.64
	√		82.89
		√	83.20
Preprocessing + Octave Convolution + KM++			90.43



(a)



(b)

Fig. 16. Curves of the training procedure on the De Gaulle airport data set for the proposed model. (a) Without the last SA. (b) With the last SA.

in our model is replaced, which would deteriorate the result. This indicates that our designed measures are very judicious and effective for the deep clustering model to produce the CDF and improve the clustering accuracy in SAR images.

The experiments of the proposed model without the last SA module versus with the last SA module are shown in Fig. 16. To analyze the issue clearly, this experiment is made on the De Gaulle Airport data set due to its more complex changed scenarios and strong level of speckle noise. As shown in Fig. 16(a), for the proposed model without the last SA, we can see that Kappa values gradually decrease as the epoch of the training increases. Inversely, the curves of the proposed model with the last SA are on the rise shown in Fig. 16(b). If the number of epochs is set larger than 50, the performance of the model is not better, and thus, we set epoch $=$ 50. In addition, the curves of the model with the last SA are not smooth, instead flexural. The reason is likely that the network and KM++ are randomly initialized, which affects the stability of the proposed model. In the unsupervised deep clustering case, the initialization is very important for the

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success of the approach. Even so, our approach still achieves the state-of-the-art results.

V. Conclusion

In this article, we propose a multiscale SA deep clustering model based on octave convolution for CD in SAR images. Our goal is to learn the CDF that is suitable for clustering with the powerful CNN in an unsupervised manner, which is important in the field of remote sensing that extremely lacks annotated data. To keep the advantages of jointly optimizing CNN and clustering, we propose an effective network model and optimization procedure to avoid trivial and meaningless clustering results and also design multiscale fusion strategies to further refine the detection result. As experiments demonstrated, the proposed model greatly boosts the performance of unsupervised CD in real SAR data sets with different resolutions. Compared with the current advanced algorithms, the proposed model shows great robustness and potential in suppressing complex backgrounds and keeping more details of changed areas. In the future work, with the goal of unsupervised difference feature learning, more suitable network architectures with good initial parameters need further study. In terms of multiscale features, a common technique is feature pyramid networks (FPNs) that exploit the inherent multiscale, pyramidal hierarchy of CNN. It will also be our next consideration to benefit from FPN to detect objects at different scales for SAR image CD.

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